

Muhammad Bilal Nazir

+92 316 7711865 | bachohan786@gmail.com | Chiniot, Pakistan
linkedin.com/in/mbilal-nazir | github.com/Bilal-Nazir-3360

SUMMARY

Computer Science graduate from FAST NUCES with hands-on experience building scalable, responsive web applications using the MERN stack, REST APIs, and modern JavaScript frameworks. I enjoy developing production-ready MERN stack applications, from AI-powered security tools to SaaS platforms and e-commerce solutions. I am a fast learner with strong attention to detail, comfortable working both independently and in collaborative teams to deliver clean, maintainable, and impactful code.

EDUCATION

Bachelor of Science in Computer Science

2022 – 2026

– Fast National University of Computer and Emerging Sciences, Chiniot-Faisalabad Campus

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, C++, Java

Frontend: HTML5, CSS3, React.js, TailwindCSS, Material-UI, Redux, Context API, Responsive Web Design

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Mongoose, Microservices

Databases: MongoDB, MySQL, PostgreSQL

ML / AI Integration: TensorFlow, PyTorch, Scikit-learn, XGBoost, NLP, GANs, OpenAI API

Tools & Other: Git, GitHub, VS Code, Postman, Jupyter Notebook, System Design, Agile, Docker

RESEARCH EXPERIENCE

Next-Gen Cyber-Attack Detection | Undergraduate Research Project

- Designed and developed a scalable, responsive AI security platform using React.js, Node.js, Express.js, Python, and PostgreSQL. Built a real-time adversarial attack generation and evaluation dashboard integrating GAN-based models with ML and signature-based Intrusion Detection Systems. Implemented continuous lifelong learning for automated IDS retraining and improvement, with a live monitoring interface for attack simulation and detection performance tracking.

PROJECTS

PhishGuard: AI-Powered Phishing Detection System

- Developed a production-grade full-stack security application for detecting phishing URLs and emails using machine learning and NLP. Achieved 92-98% URL accuracy and 90-97% email accuracy using XGBoost, SVM, and DistilBERT models. Built a responsive React.js frontend featuring real-time confidence scoring and suspicious word highlighting for explainability. Developed RESTful backend using Node.js and FastAPI with user authentication, scan history, and role-based access control.

Stellar Web Manager: Team Collaboration & AI Dashboard

- Built a SaaS-style team collaboration platform using React (TypeScript) and TailwindCSS on the frontend, with Node.js, Express.js, and MongoDB on the backend. Implemented JWT-based authentication, project and task management, real-time notifications, and analytics dashboards. Integrated OpenAI and Gemini APIs to provide an embedded AI chat assistant for productivity enhancement. Designed with a modular, scalable architecture to support enterprise-level usage.

BookVerse: Online Bookstore & Ordering Platform

- Developed a full-stack online bookstore using React.js and Node.js with a complete e-commerce workflow including browsing by category, favorites, shopping cart, and order placement. Implemented secure user authentication with JWT and built a comprehensive admin panel for managing books, categories, orders, and inventory. Designed a clean and responsive UI focused on user experience and ease of navigation.

ShopHub: MERN E-Commerce Application

- Built a full-stack e-commerce web application using the complete MERN stack — MongoDB, Express.js, React.js, and Node.js. Features include user authentication, product catalog management, shopping cart, order processing, and a clean Material-UI interface. Followed RESTful API design principles with secure backend routes and efficient database queries for optimal performance.

ACTIVITIES

Hackathon Participant | Teaching Assistant | MLSA Member | GDSC Member | Undergraduate Researcher | FYP Team Lead